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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

2017.

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

marks

1. Define stack.
2. How overflow occurs in subtraction?
3. What is the branch prediction buffer?
4. List down the functions to be performed by a serial I/O interface.
5. Define interrupt.
6. What is multithreading?
7. What is the major limitation of pipeline technique?

parallel

8. How DMA can improve I/O speed?
9. Distinguish between hardwired and micro programmed control.
10. What is superscalar operation?

Ans:

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Discuss the various techniques to represent instructions in a computer system.

Or

12. What is the need for addressing in a computer system? What are the different addressing modes available?

UNIT II

13. Brief the Intel IA-32 Pentium Architecture with the diagram.

Or

14. Explain the various addressing modes and the registers of IA-32 Pentium architecture.

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UNIT III

15. Explain the use of interrupts in the operating system with suitable examples.

Or

16. What is DMA? Explain the block diagram of DMA. Also describe how DMA is used to transfer data from peripherals.

UNIT IV

17. Draw different memory address layouts and brief the technique to increase the average rate of fetching words from the main memory.

Or

18. What is Virtual memory? Explain the steps involved in Virtual memory address translation.

UNIT V

19. Explain about the Multiple-Bus organization.

Or

20. State and explain the different types of hazards that can occur in a pipeline.

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**B.Tech. DEGREE EXAMINATION,
SEPTEMBER 2021.**

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

Time : Three hours

Maximum : 75 marks

SECTION A -- (10 × 2 = 20 marks)

Answer ALL the questions.

1. Draw the functional units of Computer.
2. Define stack and Queues.
3. State the IA-32 Pentium Input and output operations.
4. Give an example of shift and rotate instructions.
5. Define Interrupts.
6. List out the types of interfaces.
7. Write about the Flash memory.
8. Brief about the cache memories

9. Define data and Instruction hazards.

10. What is hard-wired control? How is it different from micro-programmed control?

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the following

- (a) Memory Location and address
- (b) Memory operations.

Or

12. Give the detail explanation about stacks and queues.

UNIT II

13. Write an IA32 assembly code for floating point addition and subtraction.

Or

14. Give the detail explanation about I/O operations.

UNIT III

15. Describe the hardware mechanism for handling multiple interrupt requests.

Or

16. Discuss Direct Memory Access Controller with example.

UNIT IV

17. Explain different types of mapping functions for cache memory.

Or

18. Draw the block diagrams of two types of DRAM and explain.

UNIT V

19. Explain the use of multiple-bus organization for executing a three-operand instruction.

Or

20. Illustrate the super scalar operations with a neat diagram.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the purpose of Accumulator?
2. What is the purpose of Memory Buffer Register (MBR)?
3. What is an instruction?
4. State the need for subroutines.
5. Define : interface.
6. What is DMA?

7. State the basic operation of cache memory.
8. What is RAM?
9. What is meant by hardware control?
10. Why the data bus is bidirectional?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. What is meant by addressing mode? Explain the different types addressing modes in detail.

Or

12. Explain about shift and rotate instructions in detail.

UNIT II

13. Explain about IA-32 instructions in detail.

Or

14. Discuss about IA-32 assembly language with examples.

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UNIT III

15. Discuss about Pentium interrupt structure in detail.

Or

16. Explain about handling multiple devices in detail.

UNIT IV

17. What is virtual memory? Explain in detail.

Or

18. Explain about semiconductor RAM memories in detail.

UNIT V

19. Explain how the instructions are executed in computer detail.

Or

20. What is meant by pipelining? Discuss in detail.

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B.Tech. DEGREE EXAMINATION, MARCH 2021.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. State the different addressing modes.
2. Write about the Shift and Rotate Instructions.
3. Define Assembly language.
4. Give an example of program flow control.
5. Discuss about enabling and disabling interrupts.
6. Draw the diagram of serial interface.
7. Bring out the difference between SRAMs and DRAMs
8. List the types of ROM.
9. What is hazards?
10. Brief about superscalar operations.

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PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All Questions carry equal marks.

11. (a) Distinguish the Multicomputer the Multiprocessor and Computer.

(b) Describe the various Functional units of

12. Discuss basic Input/output Operations in detail.

13. Explain any four addressing modes of Pentium Process with example.

14. Explain program flow control with example.

15. Analyze the use of vectored interrupts in processors. Why is priority handling desired in interrupt controllers?

16. List the uses of Interrupts in Operating system and explain Pentium Interrupt Structure with suitable example.

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17. Discuss the following

(a) Interleaving

(b) Hit rate and Miss Penalty

18. Explain how the virtual address is converted to real address in a paged virtual memory system.

19. Illustrate the design of hardwired control with necessary detail.

20. Describe the techniques for handling data instruction hazards in pipelining.

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B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

INFORMATION,
2014.

Third Semester

Computer Science and Engineering

Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013-14 Regulation)

Maximum : 75 marks

Time : Three hours Maximum : 75 marks

(3 marks)

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

Answer ALL questions.

All questions carry equal marks.

Answer ALL questions.

1. What are the functional units of a computer?
2. What is index register?
3. Define the term instruction pointer.
4. Write the use of rotate instruction with example.
5. What do you mean by polling?
6. What are the kinds of exception?

7. Define the term Latency.

8. Define (a) Hit ratio (b) Miss penalty.

9. State the different types of hazard that occur in pipeline.

10. List the stages of pipelining.

PART B — (5 × 11 = 55 marks)

Answer ALL the questions ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Discuss in detail about addressing modes.

Or

12. What do you understand by stack frame? Discuss its use in Subroutines.

UNIT II

13. Write and explain the control sequence for execution of an unconditional branch instruction.

Or

14. Describe the register structure of IA32.

UNIT III

15. Differentiate between memory mapped I/O and I/O mapped I/O and write how interrupts are enabled and disabled.

Or

16. Write short notes on Direct Memory Access.

UNIT IV

17. With a block diagram, analyze the memory hierarchy in terms of speed, size and cost.

Or

18. Describe how virtual memory address is translated.

UNIT V

19. Explain the execution of a complete instruction.

Or

20. Compare hardwired control with micro programmed control.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

Time : Three hours Maximum : 75 marks

PART A (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

- 1 Differentiate multiprocessor and multicomputer.
- 2 Define register mode instruction and absolute mode instruction.
- 3 What are memory segments? Name the segments used in IA-32.
- 4 Explain the unconditional jump instruction of IA-32.
- 5 What are vectored interrupts?
- 6 Define the term Full Handshake.

7. Write the use of virtual memory.
8. What are RAID disk arrays?
9. Write the actions needed for an instruction to be executed.
10. What are superscalar processors?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. With a neat diagram explain the functional units of a computer. 89

Or

12. Explain the shift and rotate instructions with examples.

UNIT II

13. Explain the machine instruction format of IA-32.

Or

Describe the I/O operations supported by IA-32.

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UNIT III

15. What is an exception? Explain in detail about the types of exceptions.

Or

16. Explain the concept of direct memory access.

UNIT IV

17. Describe about the types of ROM.

Or

18. Discuss about the mapping techniques of cache memory.

UNIT V

19. Explain the concept of hardwired control with neat diagram.

Or

20. What is meant by hazard? Explain the types of hazard with examples.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Bring the difference between computer architecture and computer organization.
2. Define multiprocessor.
3. Write notes on IA-32 Pentium.
4. Give the instruction format of IA-32.
5. Draw the I/O interface for an input device.
6. Define distributed arbitration.
7. Define asynchronous DRAMs.
8. What are the types of flash memory?
9. Define multiphase clocking.
10. What are the steps involved in processing the instruction in pipelined processor?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions.

UNIT I

11. Explain the basic instruction types.

Or

12. With an example, explain indirection and pointers.

UNIT II

13. Discuss various addressing modes of IA-32.

Or

14. Explain subroutines concept in IA-32 architecture.

UNIT III

15. Write notes on :

(a) Enabling and disabling interrupts (5)

(b) Handling multiple devices. (6)

Or

6. Discuss about :

(a) Pentium interrupt structure (6)

(b) Direct memory access. (5)

UNIT IV

17. Explain about mapping functions in cache memory.

Or

18. Explain address translation in virtual memory.

UNIT V

19. Explain hardwired control in detail.

Or

20. Explain conditional branches and branch prediction in detail.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013 -2014 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Mention the advantages of multiprocessors.
2. Name the basic functional units of a computer.
3. Give some addressing modes used in IA-32 processor.
4. What are the registers of IA-32 processor?
5. Define an interrupt?
6. What is DMA?
7. Give the importance of ROM.

8. Define the memory hierarchy used in computers.
9. Define data hazards.
10. Mention the services by a control unit.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. With a neat architecture diagram, briefly explain about the functional units of a computer.

Or

12. Write notes on:

- | | |
|----------------------------|-----|
| (a) Instruction sequencing | (4) |
| (b) Multicomputers | (4) |
| (c) Assembly Languages | (3) |

UNIT II

13. Classify various instructions in the instruction set of IA-32 processor

Or

14. Explain with a program example about subroutine processing of IA-32.

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UNIT III

15. Briefly explain in detail about Direct Memory Access.

Or

16. Explain about various standard I/O interfaces in detail.

UNIT IV

17. Explain the need for virtual memory. Also explain in detail about address translation in TLB with a neat diagram.

Or

18. Explain various secondary storage devices with neat diagrams.

UNIT V

19. Briefly explain with examples about how the data hazards and control hazards are handled?

Or

20. Write notes on

- | | |
|-----------------------------|-----|
| (a) Hardwired control | (6) |
| (b) Microprogrammed control | (5) |

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013 – 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is an address space?
2. What do you mean by instruction sequencing?
3. What are the conditional flags?
4. Why we need a stack?
5. Distinguish between an internal and external interrupt.
6. What is memory mapped I/O?

7. What is a virtual memory?
8. Determine the number of address lines required for 512 bytes of memory.
9. Why we need a micro program?
10. What is instruction hazard?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Give detailed account on : Addressing modes.

Or

12. Explain the different types of shift and rotate instructions.

UNIT II

13. Explain with example how to use program flow control instructions in IA-32 Processor.

Or

14. With suitable example explain how to use subroutines in IA-32 Pentium processor.

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UNIT III

15. Explain how to handle interrupts requests from multiple devices.

Or

16. What is a DMA Controller? Explain its hardware architecture with a neat diagram.

UNIT IV

17. Write notes on : Semiconductor RAM memories.

Or

18. What is Cache memory? Explain different types of mappings from main memory to cache memory.

UNIT V

19. Briefly describe the design of a hardwired control unit.

Or

20. Explain the super scalar operations.

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UNIT - V

19. With neat diagram explain the organization of a hardwired and micro programmed control unit.

Or

20. With neat diagram illustrate how each instruction are to be executed by the pipelined processor

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B.Tech DEGREE EXAMINATION, MAY 2019.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013 - 14 onwards Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions

All questions carry equal marks.

1. List out the functional units of a computer.
2. Name the two registers which facilitate communication between the memory.
3. Define IA-32 instruction format.
4. Name the I/O instructions used in IA-32.
5. Differentiate interrupt and exception.
6. Differentiate synchronous and Asynchronous bus.

7. Differentiate memory access and cycle time.
8. State the uses of cache memory.
9. What are the necessary steps to be followed to execute an instruction?
10. List out various hazards that cause performance degradation in pipelined processors.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT - I

11. Illustrate various generic addressing mode with an example.

Or

12. With an example illustrate how to apply shift and rotate instructions on general binary-coded information

UNIT - II

13. (a) With neat diagram explain IA-32 register structure. (6)
- (b) Write an IA-32 code for the byte sorting program. (5)

Or

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14. (a) In IA-32 terminology explain branch instructions (both conditional and unconditional) with an example. (6)
- (b) Write an IA-32 subroutine for deleting record from a linked list. (6)

UNIT - III

15. With neat diagram explain the Pentium interrupt structure. (6)

Or

16. State the benefits of Direct Memory Access (DMA) and explain its bus arbitration mechanism. (6)

UNIT - IV

17. (a) With neat diagram explain in detail the organization of a 1K X 1 memory chip. (5)
- (b) With neat diagram explain the internal organization of a 2M X 8 dynamic memory chip. (6)

Or

18. With neat diagram explain virtual memory address translation mechanism. (6)

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B.Tech. DEGREE EXAMINATION,
MAY 2018.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is an instruction cycle? Give examples of instructions with different number of instruction cycles.
2. List the different types of computer registers.
3. What is program control?
4. Differentiate between floating point and decimal arithmetic operations.
5. What is the use of Cache memory?
6. Why Virtual memory is required?
7. What are the functions of input output interface?

8. Distinguish between the different modes of data transfer.
9. What are the different types of multiple processor organizations?
10. What are the advantages of non-uniform memory access?

PART B — (5 × 11 = 55 marks)
Answer ALL questions, ONE from each Unit.
All questions carry equal marks.

UNIT I

11. Describe different types of computer instructions with examples for each.
- Or
12. Explain the design of basic computer and accumulator logic.

UNIT II

13. What are the different addressing modes? Explain with examples.
- Or
14. With suitable diagrams explain register organization and stack organization.

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15. Explain any three types of memory.

UNIT III

Or

16. Compare and contrast different types of memory.

UNIT IV

17. What is cycle stealing? Explain how they are used in any two instances in the working of a computer.

Or

18. What are the different types of interrupt? With suitable examples explain how they are used in both system and application programs.

UNIT V

19. Explain how Cache coherence could be achieved.
- Or
20. Describe Vector Computation.

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B.Tech. DEGREE EXAMINATION, JANUARY 2022.

Third Semester

Computer Science And Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013 – 2014 Regulations)

Time : Three hours

Maximum : 75 marks

SECTION A – (10 × 2 = 20 marks)

Answer ALL questions.

1. Write any two application of Multiprocessor.
2. Define Memory Address Register.
3. What is the functions of the register in Intel architecture 32?
4. Mention the purpose of shift and rotation instructions.
5. Why IO devices cannot be directly connected to the system bus?
6. Define DMA Controller.

7. What are the enhancements used in the memory managements?

8. What is Cache memory?

9. Define Hardwired Control unit.

10. Which technique is used to solve the data hazard?

SECTION B – (5 × 11 = 55 marks)

Answer ALL questions

UNIT I

11. Explain the functional units of a computer with a neat sketch.

Or

12. Explain different types of instructions with examples. Compare their relative merits and demerits.

UNIT II

13. Explain Instruction Execution Cycle with the sequence of steps in Pseudo code format.

Or

14. Illustrate the various IA-32 registers with neat sketch.

UNIT III

15. Describe Interrupt and its types with examples.

Or

16. What are the handling of multiple devices? Explain the methods that are used to decide the device to select.

UNIT IV

17. Explain about the direct and set associative replacement technique in cache.

Or

18. What are the factors affecting computer performance? Explain in detail.

UNIT V

19. Discuss the principles operations of the microprogrammed control unit.

Or

20. (a) Give a brief note on Data path.

(b) List the advantages and disadvantages of superscalar architecture.

UNIT IV

- (a) Explain the organization and accessing of data on a Disk. (7)
- (b) Explain how the virtual address is converted into real address in a paged virtual memory system. (4)

Or

Explain the various mapping techniques associated with cache memories with example.

UNIT V

Describe the techniques for handling data and instruction hazards in pipelining.

Or

Explain the basic organization of a microprogrammed control unit and the generation of control signals using microprogram.

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B.Tech. DEGREE EXAMINATION, APRIL, 2015.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND
ARCHITECTURE

(2013-2014 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Distinguish between auto increment and auto decrement addressing modes.
2. What is SPEC? Specify the formula for SPEC rating.
3. What is an opcode? How many bits are needed to specify 32 distinct operations?
4. State the addressing modes supported by IA-32 Pentium processor.
5. What is meant by bus arbitration?

6. How are interrupt requests from multiple devices handled?
7. How many memory chips are needed to construct a $2\text{ M} \times 16$ memory system using $512\text{ K} \times 8$ static memory chips?
8. List the factors that determine the storage device performance.
9. What is hard-wired control? How is it different from micro-programmed control?
10. What is meant by hazards in pipelining?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain the different types of addressing modes with suitable example. (6)
 - (b) What are the special registers in a typical computer? Explain their purpose with examples. (5)
- Or
12. (a) Describe the organization of stack. (4)
 - (b) Explain with a diagram the design of a fast multiplier using carry save adder circuit. (7)

UNIT II

13. (a) Explain in detail about the various addressing modes available in IA-32 Pentium processor with example.
 - (b) Write an assembly code to check whether data A is positive or negative number.
- Or
14. (a) Explain in detail about the Instruction Queue of IA-32 Pentium processor.
 - (b) Write an assembly code to check whether data X is odd parity or even parity data.

UNIT III

15. (a) Explain the protocols of USB with diagram.
 - (b) Describe the hardware mechanism for handling multiple interrupt requests.
- Or
16. (a) What are handshaking signals? Explain handshake control of data transfer input and output operation.
 - (b) Explain in detail about Pentium interrupt structure.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Computer Science and Engineering

COMPUTER ORGANIZATION AND ARCHITECTURE

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are Multiprocessors?
2. Define Addressing mode? What are subroutines?
3. What are the Registers available with IA-32?
4. Give example for Shift/Rotate instructions?
5. What do you mean by an interrupter?
6. What are the advantages of Direct Memory Access?

UNIT III

7. What are the various types of ROM?
8. Define virtual memory.
9. Write differences between Hardwired and microprogrammed controls.
10. How can we solve data Hazards in pipelining?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each unit.
All questions carry equal marks.

UNIT I

11. Explain the various types of addressing mode with example.

Or

12. Explain the shift and rotate operations with suitable example.

UNIT II

13. Explain the architecture of IA-32 with suitable Block diagram.

Or

14. Explain the I/O operations IA-32 with example.

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UNIT III

15. Explain the interrupt enabling and disabling mechanism.

Or

16. Explain the process of DMA with block diagram.

UNIT IV

17. Explain the memory hierarchy with respect to speed, size and cost.

Or

18. Explain the concept of virtual memories.

UNIT V

19. Give note on

- (a) Complete Instruction Execution
- (b) Microprogrammed control.

Or

20. What are data and Instruction hazards and how can this be avoided?

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UNIT V

19. Consider un-pipelined machine with 10 ns clock cycles. It uses 4 cycles for ALU operations and branches whereas five cycles for memory operations. Assume that the relative frequencies of these operations are 40%, 20% and 40% respectively. Suppose that due to clock skew and setup, pipelining the machine adds 1 ns overhead to the clock. Calculate gain in the instruction execution rate from a pipeline.

Or

0. Discuss the following terms :
- (a) Cache structure in Pentium
 - (b) Brief Super scalar architecture
 - (c) What are the advantages of hardwired control unit? What are the disadvantages?

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

Computer Science and Engineering
COMPUTER ORGANIZATION AND
ARCHITECTURE

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Brief zero address instructions format.
2. Define the fundamental conceptual unit in a computer.
3. Compare the features of P-V and P-VI in brief.
4. Distinguish I/O instructions and I/O commands.
5. The interrupt vector table is always created in the first 1K area of the memory. Justify the statement.
6. Define interface circuit. List any two I/O interfaces and its purpose.
7. Briefly explain the significance of DDRAM.

8. Explain the need of overlaying concept in memory management.
9. Distinguish RISC and CISC architecture.
10. Brief data hazards and instruction hazards in pipelining techniques.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Define subroutine. Explain subroutine linkage using a link register. (6)
- (b) Brief (i) byte addressability (ii) big-endian assignment (iii) little-endian assignment. (5)

Or

12. (a) With a neat sketch, explain single bus and multiple bus structure used to inter-connect functional units in computer system. (6)
- (b) Explain how the performance of a computer can be measured. List the measures that are used to improve the performance of computers. (5)

UNIT II

13. (a) What is memory paging? Explain how it is used for memory addressing in Pentium processor. (6)
- (b) Write an assembly language program to find one's complement and two's complement of an 8-bit number. (5)

Or

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14. (a) What features have been added to P-IV improve Paging?
- (b) Find greatest of five numbers and the GC for the same five numbers using assembly language program.

UNIT III

15. Explain about the following Input-Output interface with an example :
 - (a) I/O bus and interface modules
 - (b) Isolated versus memory mapped I/O
 - (c) I/O versus memory bus.

Or

16. With a neat sketch, explain the Pentium interrupt structures.

UNIT IV

17. Explain the following memory organization methods in virtual memory with diagrams :
 - (a) Address space and memory space
 - (b) Address mapping using pages
 - (c) Associative memory page table
 - (d) Three benefits of virtual memory

Or

18. (a) Explain in detail the various memory management techniques.
- (b) Explain the various modes in which DMA works.

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